RelCompat3D: Re-Ranking 3D Scene Graph Relations with Geometric Evidence Supplementary Material

Anonymous submission

A Supplementary Method Details

Overview. Section A gives the notation, compatibility inputs, target construction, estimators, and formal properties. Section B records the data, preprocessing, training environment, and internal regeneration check. Section C provides the additional results referenced by the main paper. Section D collects secondary diagnostics that bound robustness and scope.

A.1 Notation

Table S1 summarizes the notation used in the main paper and the supplement.

Symbol	Meaning
r_i	Relation candidate i
(s_i, o_i)	Ordered subject and object instance identifiers
$p_i,\ a_i$	Predicate label and relation-family label
$k_i^{\text{pair}},\ k_i^{\text{rel}}$	Ordered-pair identity and exact relation-candidate identity
$T_i,\ G_i,\ Z_i$	Predicate semantics, ordered-pair measurements, and source relation score
$q,\ f_q,\ C_i^q$	Compatibility estimator, compatibility logit, and bounded compatibility score
$H_{a_i},\ \mathcal{O}_i$	Applicable transformation group and transformation orbit
$C_i^{\text{tr,q}},\ u_i^q$	Transformation-consistent compatibility and within-family ranking score
$\mathcal{A}_{\text{eval}},\ \mathcal{A}_{\text{rank}}$	Evaluated and re-ranked relation-family sets
$\pi^Z,\ \pi_a^q$	Source ranking and ordered list for family a
$L_K(c),\ Y_c$	Selected top-K candidates and exact ground-truth relations in context c
$D_K,\ N_s, N_u, N_v$	Status-assigned candidates and satisfied, uncertain, and violated counts
$\mathcal{P},\ \ell_i^q$	Linked training pairs and compatibility logit used by the pairwise loss

Table S1: Notation used in the RelCompat3D method and evaluation.

A.2 Target Construction and Information Boundary

Ground-truth relations in the evaluated families provide positives. Negatives are generated within the same scene context and linked to their source positive. The construction does not use source relation scores, predictor identities, or evaluation candidates. Table S2 gives the complete construction policy.

Information-use boundary. Table S3 makes the information boundary explicit. Training examples come only from the training split, so evaluation candidate rows, source relation scores, and primary verifier-status labels do not enter target construction. The counterfactual rules and primary verifier nevertheless share some OBB-derived measurements and family thresholds. The point- and mesh-based audit removes that OBB overlap but retains the reconstructed evaluation scenes and relation ontology.

We cap negatives at two per positive, 200 per context–family, and three times the number of positives per family. These caps limit repeated easy negatives but do not equalize family counts. The constructed training and internal-development sets together contain 28,977 positives and 37,477 negatives. Their combined family totals are 23,008 proximity, 5,488 vertical, and 37,958 support/contact rows. The explicit overlap above is why Violation computed by the rule-based geometry verifier is reported as a verifier-derived measure rather than independent physical-validity ground truth. The feature-removal analysis tests this dependence but does not eliminate it.

A.3 Estimator Architecture and Optimization

RelCompat3D-Linear fits one logistic model per evaluated relation family. Numeric features use means and standard deviations estimated from the training split. Missing values use the corresponding training mean. Inputs are predicate indicators T, predicate-independent geometric measurements of the ordered pair G, and the two predicate-signed height interactions defined in the main paper. The family label selects the head and is not repeated as an input. The source relation score, source rank, predictor identity, and object class are excluded. Transformation-consistent augmentation preserves the proximity predicate under endpoint swap. Vertical order swaps endpoints and applies the inverse predicate. Linked counterfactuals receive the fixed margin loss in the

Family	Construction rule	Relation to verifier
Proximity	Choose directed same-context pairs in descending normalized XY distance. Require distance ≥ 2.5 and zero projected overlap.	Shares distance/overlap primitives and the 2.5 satisfied boundary. The verifier declares violation only at ≥ 3.5.
Vertical	On the same directed pair, replace `higher than` with `lower than` or conversely. Require the positive direction to have $\lvert\Delta z\rvert \geq 0.25$ m and normalized $\lvert\Delta z\rvert \geq 0.15$.	Shares both signed primitives and thresholds.
Support/contact	Replace the subject or object within the context. Reject known support/contact pairs in either direction and floor subjects. Require any two of normalized XY distance ≥ 2.0, zero overlap, and vertical gap ≥ 0.30 m.	Shares these OBB primitives and thresholds. Point-subtype evidence used by the reported verifier is not used to generate negatives.

Table S2: Counterfactual-negative construction. Normalized XY distance uses the mean ordered-pair XY diagonal. Zero overlap requires both directed projected-overlap ratios to be at most 10^{-9}.

Information	Train	Verifier	Audit
Evaluation candidates	No	Yes	Yes
Source score	No	No	No
Verifier labels	No	Output	No
OBB measurements	Some	Yes	No
Point/mesh measurements	No	No	Yes
Evaluation scene identities	No	Yes	Yes
Relation ontology	Yes	Yes	Yes

Table S3: Information used by counterfactual training construction (Train), the primary verifier (Verifier), and the point/mesh audit (Audit). "Some" denotes the OBB primitives and family thresholds shared by training construction and the primary verifier. The audit uses alternative measurements, not independent physical-validity ground truth.

main paper. Support/contact receives the pairwise loss but no blanket endpoint transform.

Geometric feature specification. Table S4 defines the 17 measurements in G. For endpoint $j \in \{s, o\}$, let $c_j = (x_j, y_j, z_j)$ be the OBB center, b_j and t_j its lower and upper scene-coordinate bounds, $h_j = t_j - b_j$, and d_j^{3D} and d_j^{XY} the diagonals of its 3D and projected XY bounds. Let A_j be the area of the projected bounds and I_{XY} their intersection area. All distance and height measurements are in meters, and normalized and overlap measurements are dimensionless.

RelCompat3D-MLP fits one shared multilayer perceptron (MLP) with a single two-unit ReLU hidden layer and a predicate-linear skip connection. Its 29 inputs comprise family and predicate indicators, the same 17 geometric measurements, the two signed-height interactions, and the sum of the directional overlap ratios. The family indicator is nonconstant because this estimator shares parameters across families. It has 69 parameters and excludes the source relation score and predictor identity.

The 60,208 training rows yield 33,961 linked pairs, and transformation-consistent augmentation produces 86,032 optimization rows for both estimators. Linear uses 800 deterministic full-batch gradient-descent steps at learning rate 0.2. MLP uses 120 full-batch Adam steps at learning rate 0.02. Both use the same pairwise weight, margin, ℓ_2 coefficient, and fixed seed policy. The three linear family models contain 21–23 parameters. Each reported condition uses one fit with a fixed seed. The paired intervals quantify scan resampling rather than variation across training runs.

A.4 Formal Properties

Proposition 1 (Exact transformation consistency) *Let a finite transformation group H_a act jointly on the predicate and geometry of an ordered pair, and define $C_a^{\mathrm{tr}}(x) = |H_a|^{-1} \sum_{g \in H_a} C_a(gx)$. Then $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$ for every $h \in H_a$.*

Proof. For any $h \in H_a$, $C_a^{\mathrm{tr}}(hx) = |H_a|^{-1} \sum_{g \in H_a} C_a(ghx)$. Right multiplication by h is a bijection on the finite group, so re-indexing $g' = gh$ gives $C_a^{\mathrm{tr}}(hx) = C_a^{\mathrm{tr}}(x)$. The two-element proximity and vertical actions yield the identities in the main paper, and the singleton support/contact action is the identity.

Family-sequence preservation. For each family a, the algorithm constructs the ordered list π_a defined in the main paper. At position j, it selects the next unselected candidate from π_{a_j}. The output family sequence is therefore exactly $(a_1, \ldots, a_n)$. The support/contact list follows the source ranking, so the first K positions contain exactly the same ordered support/contact subsequence as the source prefix. Therefore the support/contact selections and their order are identical at every reported K.

Prefix utility optimality. Fix a prefix length K, and let $n_a(K)$ be the number of positions assigned to family a in the source prefix. Every feasible ranking that preserves these family counts must select $n_a(K)$ candidates from each family. Because each re-ranked π_a is sorted by u_i, each estimator selects the $n_a(K)$ highest-scoring candidates in that family. Replacing one of these candidates with a lower-scoring candidate from the same family would reduce the summed score of the prefix. The support/contact subsequence is fixed, so it does not affect this comparison.

B Reproducibility and Experimental Setup

B.1 Data, Predictors, and Evaluation Universe

Compatibility fitting uses 1,061 training scans. Model design and relation-family choices use a disjoint 117-scan de-

Feature	Definition
`distance_3d`	$\|c_s - c_o\|_2$
`distance_xy`	$\sqrt{(x_s - x_o)^2 + (y_s - y_o)^2}$
`normalized_distance_3d`	$2\,\texttt{distance\_3d}/(d_s^{3D} + d_o^{3D})$
`normalized_distance_xy`	$2\,\texttt{distance\_xy}/(d_s^{XY} + d_o^{XY})$
`center_delta_z`	$z_s - z_o$
`normalized_center_delta_z`	$2(z_s - z_o)/(h_s + h_o)$
`projected_iou_xy`	$I_{XY}/(A_s + A_o - I_{XY})$
`projected_subject_overlap_ratio`	I_{XY}/A_s
`projected_object_overlap_ratio`	I_{XY}/A_o
`vertical_gap_subject_on_object`	$b_s - t_o$
`subject_bottom_z`, `subject_top_z`	$b_s,\ t_s$
`object_bottom_z`, `object_top_z`	$b_o,\ t_o$
`abs_center_delta_z`	$\|z_s - z_o\|$
`abs_normalized_center_delta_z`	$\|2(z_s - z_o)/(h_s + h_o)\|$
`abs_vertical_gap_subject_on_object`	$\|b_s - t_o\|$

Table S4: The 17 predicate-independent geometric measurements in G. Projected areas use scene-coordinate XY bounds recovered from the annotation OBBs. A missing numeric value is replaced by its training-split mean before standardization.

velopment split. The shared evaluation universe contains 157 official 3DSSG validation scans, 548 relation contexts, and 3,972 exact-match ground-truth relations in the support/contact, proximity, and vertical-order families. Evaluation candidates, source relation scores, and verifier-status labels are not used for fitting. The same fitted estimators are applied to all three predictors without predictor-specific refitting.

VL-SAT. We rerun the released VL-SAT Mmgnet checkpoint on the official validation scenes with its 160-object and 26-relation ontology. Evaluation disables data augmentation and uses 128 sampled points per object and 256 per union. The adapter applies a sigmoid to each relation logit and exports all 26 non-none predicate scores for every ordered pair. This produces 957,008 candidates over 36,808 ordered pairs and 548 contexts. We retain the six predicates in the three evaluated families and join each candidate to geometry by its scan, context, subject, and object identifiers.

Open3DSG. We train Open3DSG using the official implementation and non-averaged BLIP configuration, and select the checkpoint using training and development losses only. Public preprocessing produces candidates for 533 of the 548 contexts. The main evaluation retains all 548 contexts and treats the 15 contexts without candidates as empty sets. An eligible-context sensitivity analysis uses 3,899 relations. A separate recovery covers all 548 contexts by lowering the minimum number of visible objects to two and regenerating views for two scans.

SGFN. We evaluate the released SGFN checkpoint with its 160-object and 26-relation configuration on the same validation scenes and denominator.

Source relation scores. The observed proximity and vertical-order score ranges are $[5.30\times10^{-22}, 0.9954]$ for VL-SAT, $[0.6394, 0.9281]$ for Open3DSG, and $[4.61\times10^{-20}, 0.5846]$ for SGFN. All evaluated scores are positive, so multiplication by compatibility cannot reverse their order through a sign change. These ranges are observations, not calibrated probability intervals. The Open3DSG range over all relation families is $[0.5772, 0.9707]$.

B.2 Environment and Internal Validation

Implementation and development diagnostics. All RelCompat3D preprocessing, fitting, and evaluation jobs use a pinned Docker image based on Python 3.11.9 with NumPy 1.26.4. Re-ranking requires no GPU. Source-predictor inference runs in separate source-specific containers and is excluded from the timing below. Compatibility is computed once per candidate and cached before re-ranking. Across five timed runs, compatibility scoring, transformation averaging, and family-aware sorting took 1.81–2.45 seconds per predictor on an Intel Core Ultra 7 265KF CPU.

Development-split checks did not select predictor-specific models. Both estimators satisfied the proximity and vertical-order transformation identities to numerical precision. The MLP ordered 99.23% of linked positive–counterfactual development pairs correctly. The paper claims are based on the fixed validation protocol.

Internal row-level regeneration check. We created a deterministic set of derived rows for the 548-context evaluation universe. It contains pseudonymized candidate and context identities, predictor, predicate and family labels, source relation score, Linear and MLP compatibility, paper ranking positions, exact-match membership, and primary and point- and mesh-based verifier statuses. Original scan and instance identifiers, object categories, and raw geometric measurements are excluded. From these rows, one Docker command regenerates Tables 1–3 and the data for Figure 3. It also creates a verification rendering and compares every cell with the reported paper values. The completed run checks 291 cells. The maximum absolute error is zero under the 10^{-12} tolerance. The row schema excludes licensed scene data and direct source identifiers. The rows can be regenerated from licensed external inputs.

C Additional Results Supporting Main Claims

C.1 Estimator Components and Feature Removal

Matched component diagnostics. Tables S5–S6 isolate the linked pairwise loss and transformation averaging for both compatibility estimators. Within each estimator, all conditions retain the same rows, features, optimizer, source relation scores, and family-aware re-ranking procedure. The No pairwise condition removes only the linked pairwise term. It retains relation-preserving augmentation and transformation averaging. The No averaging condition uses the fitted full estimator without inference-time transformation averaging.

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Linear full	.9277/.0197	.9658/.0295
	Linear no pairwise	.9277/.0198	.9658/.0297
	Linear no averaging	.9277/.0197	.9660/.0295
	MLP full	.9272/.0189	.9650/.0296
	MLP no pairwise	.9275/.0180	.9648/.0280
	MLP no averaging	.9275/.0189	.9645/.0296
Open3DSG	Linear full	.4418/.0342	.5685/.0324
	Linear no pairwise	.4418/.0342	.5692/.0324
	Linear no averaging	.4413/.0342	.5692/.0324
	MLP full	.4670/.0413	.5989/.0371
	MLP no pairwise	.4600/.0399	.5939/.0362
	MLP no averaging	.4507/.0398	.5735/.0368
SGFN	Linear full	.7450/.0263	.9303/.0350
	Linear no pairwise	.7450/.0264	.9303/.0353
	Linear no averaging	.7450/.0263	.9303/.0350
	MLP full	.7457/.0258	.9288/.0350
	MLP no pairwise	.7452/.0255	.9295/.0330
	MLP no averaging	.7457/.0258	.9300/.0352

Table S5: Matched component removals. Cells are exact-match Recall / verifier-derived Violation reported as proportions. Full, no-pairwise, and no-averaging conditions are matched separately within each estimator.

The pairwise term raises the held-out positive-win rate by 0.085 percentage points for Linear and 0.057 points for MLP. It also increases the mean Linear margin and reduces its softplus margin loss from .03607 to .03534. The MLP margin summaries are mixed: its mean margin and softplus loss do not improve. The term is therefore treated as a training regularizer with limited, estimator-dependent diagnostic effect, not as the main cause of the aggregate Recall–Violation changes.

Transformation averaging gives zero compatibility error and identical top-K membership under transformed representations for both full estimators and their no-pairwise refits. Without averaging, the minimum top-K Jaccard is .9635 for Linear and .6975 for MLP. Thus averaging supplies the claimed exact consistency guarantee even though its aggregate effect in Table S5 is small.

Table S7 reports the feature-removal results under the same family-aware re-ranking procedure.

Every feature-removal model is refitted on the 1,061 training scans with the same linked positive–counterfactual objective, exclusion of the source relation score, and exact transformation averaging. At $K = 50$, the strict conditions retain a clear Open3DSG improvement but not the full Violation reduction on VL-SAT or SGFN. At $K = 100$, all feature-removal conditions improve both reported metrics across all three predictors. These results show that performance does not depend on one verifier input alone. They do not eliminate the broader dependence on correlated geometry. The main paper reports matched controls under family-aware re-ranking. The pooled family-conditioning result is summarized below.

C.2 Alternative-Geometry Audit

The primary verifier and compatibility estimators share some OBB-derived measurements. We therefore construct an alternative audit from reconstructed instance points and meshes. Vertex-based distances and heights provide the point estimate. Area-weighted samples from reconstructed mesh triangles provide the mesh estimate. A candidate is labeled satisfied or violated only when these estimates agree. A disagreement is labeled uncertain. Thresholds are fixed from training-split positives. Neither estimate uses OBB measurements, source scores, learned compatibility, predictor identity, or primary verifier labels.

Table S8 reports the agreement-based Violation values for both estimators at all five K values.

For each estimator, agreement-based Violation is lower than or tied with Source in all 15 predictor–K settings. The only tie is SGFN at $K = 5$. The paired 95% interval is below zero in every non-tied setting. Point-only and mesh-only point estimates have the same non-increasing direction. At $K = 50$, measured coverage ranges from 95.5% to 98.3%, and decidable coverage ranges from 71.0% to 85.7% across predictors and estimators.

Synthetic point and mesh interventions further check the audit implementation. Translating one object laterally changes proximity labels as expected, and reversing the relative heights changes vertical-order labels. These checks are implementation diagnostics rather than additional evidence of physical validity.

C.3 Qualitative Pair Analysis

The Open3DSG context used for the main-paper teaser also contains a complementary promotion case. The exact-label relation “desk close by chair” is verifier-satisfied, with an XY center distance of $0.436\,\mathrm{m}$. RelCompat3D-Linear moves this candidate from source rank 81 to rank 30, bringing it into the top-50. This case records a promotion outcome without implying a direct exchange with the vertical-order candidate, because the two candidates belong to different relation families.

Figure S1 links each ordered-pair projection to the measured evidence and ranking outcome. The proximity and vertical-order examples are demoted by both compatibility estimators. The support/contact example is retained in source order because that family is outside the re-ranking scope.

Estimator	Condition	Family	Mean	P95	Max	Min Jaccard	Min. exact-context rate
Linear	Full	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
Linear	No averaging	Proximity	.0150	.0817	.4532	.9635	.8266
Linear	No averaging	Vertical	.0014	.0080	.0192		
MLP	Full	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No pairwise	Both	.0000	.0000	.0000	1.0000	1.0000
MLP	No averaging	Proximity	.1684	.5805	.9370	.6975	.4617
MLP	No averaging	Vertical	.0402	.2896	.9157		

Table S6: Transformation diagnostics. Mean, P95, and Max are the worst source-specific summaries of the absolute compatibility difference under the applicable endpoint/predicate transformation. The last columns give the minimum transformed-view top-K membership consistency over all predictors and reported K.

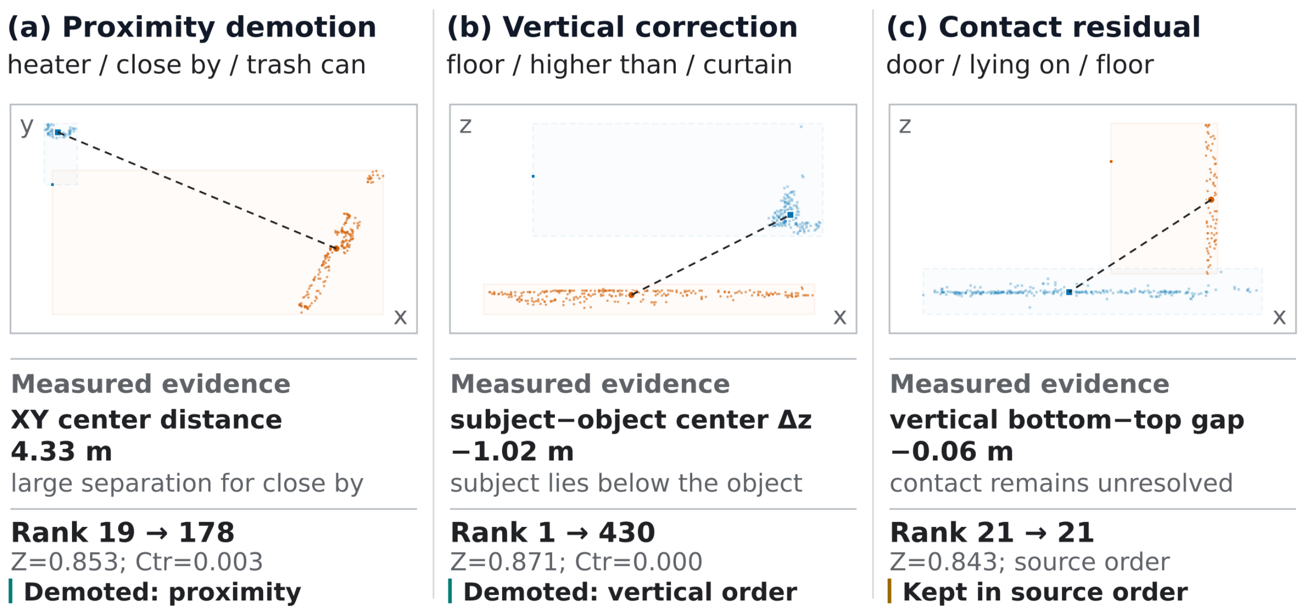

Figure S1: Pair–evidence–outcome analysis. Orange circles and solid boxes identify the subject. Blue squares and dashed boxes identify the object. The first two violations are demoted. The support/contact example requiring contact evidence is kept in source order.

C.4 Score, Baseline, and Routing Sensitivities

Source-score mapping sensitivity. The source ranking is invariant to a strictly monotonic score mapping, whereas the product $Z_i C_i^{\mathrm{tr},q}$ need not be. We therefore apply a fixed grid of mappings before product scoring without selecting a replacement mapping from evaluation results. The smooth grid contains

$$g_\gamma(Z) = Z^\gamma, \quad \gamma \in \{0.5, 2, 4\},$$

and

$$g_\tau(Z) = \sigma\big(\mathrm{logit}(\bar{Z})/\tau\big)\,, \\ \bar{Z} = \mathrm{clip}(Z, \epsilon, 1-\epsilon), \\ \epsilon = 10^{-6}, \qquad \tau \in \{0.5, 2\}.$$

The temperature mappings are monotonic scale stresses and do not treat the Open3DSG cosine score as a calibrated probability. A separate context-and-family percentile mapping assigns descending scores from one to zero within each context and family.

Table S9 summarizes all five reported K values. A setting meets both criteria when Recall is no lower and verifier-derived Violation is no higher than Source. All 75 Linear settings and 74 of the 75 MLP settings meet both criteria under the smooth mappings. The only exception is VL-SAT MLP at $K = 50$ under Z^4, where Recall changes by -0.025 percentage points, its paired interval includes zero, and Violation decreases. The percentile stress produces two Linear and four MLP Recall losses, none larger than 0.227 percentage points, without increasing Violation. Across the complete grid, including percentile mappings, the minimum pair-weighted Kendall correlation with identity-product ranking is 0.810 and the minimum $K = 50$ micro-Jaccard overlap is 0.815. These results show limited sensitivity over the tested mappings, not score-scale invariance.

Predictor	Condition	$K = 50$	$K = 100$
VL-SAT	Source	.9272/.0268	.9635/.0476
	Full model	.9277/.0197	.9658/.0295
	Exact verifier scalar removed	.9280/.0199	.9660/.0300
	Related measurements removed	.9275/.0268	.9653/.0468
	Alternative evidence only	.9280/.0268	.9642/.0471
Open3DSG	Source	.4043/.1387	.5111/.1242
	Full model	.4418/.0342	.5685/.0324
	Exact verifier scalar removed	.4381/.0342	.5687/.0325
	Related measurements removed	.4119/.0954	.5264/.0955
	Alternative evidence only	.4104/.1194	.5327/.1188
SGFN	Source	.7402/.0385	.9235/.0630
	Full model	.7450/.0263	.9303/.0350
	Exact verifier scalar removed	.7465/.0268	.9300/.0358
	Related measurements removed	.7467/.0388	.9298/.0616
	Alternative evidence only	.7462/.0386	.9277/.0625

Table S7: Feature-removal analysis. Cells are exact-match Recall / verifier-derived Violation under the same family-aware re-ranking procedure. Exact verifier scalar removed omits the normalized scalar consumed by the verifier. Related measurements removed omits all directly related distance or vertical measurements. Alternative evidence only retains projected overlap for proximity and horizontal-distance/overlap context for vertical order.

Closest simple baseline. The non-learned robust-density baseline is fitted only from training-split positive geometry. It uses predicate-specific medians and interquartile ranges, transformation averaging, the same product score, and the same family-aware re-ranking rule. It does not use verifier labels from the evaluation or constructed counterfactual negatives. Table S10 reports all K values. At $K = 50$, both learned estimators have higher Recall and lower Violation than this baseline for all three predictors. Over all 15 predictor–K settings, Linear improves both metrics in 12 settings and trades one metric against the other in three. MLP improves both metrics in 12 settings and has a trade-off in two. The robust-density baseline improves both metrics over MLP only for Open3DSG at $K = 5$.

Compatibility estimators and matched fusion. The MLP is a matched nonlinear estimator rather than a separate method. RankAvg and reciprocal rank fusion (RRF) use the Linear compatibility ranks:

$$\begin{aligned} \mathrm{RankAvg}_i &= \tfrac{1}{2}\left(\widetilde{r}_i^Z + \widetilde{r}_i^C\right), \\ \mathrm{RRF}_i &= \frac{1}{60 + r_i^Z} + \frac{1}{60 + r_i^C}. \end{aligned} \tag{S1}$$

Here r_i^Z and r_i^C are within-family ranks, and tildes denote normalized ranks. Main Table 1 reports both fusion rules at every K. Their predictor-dependent Recall–Violation trade-offs show that the result is not explained by replacing the product with a uniformly stronger fusion rule.

Matched structural controls. Table S11 reports the matched structural controls for the MLP estimator. Main Table 2 reports the corresponding Linear controls.

Across both reported K values and both compatibility estimators, corrupted predicates and ordered-pair measurements increase verifier-derived Violation. Compatibility-only ordering remains close to the full estimator on Open3DSG but loses substantial Recall on VL-SAT and SGFN. These controls support the predictor-dependent role of the source relation score described in the main paper.

Routing-constraint controls. The *family-aware* rule preserves the source family label at every rank position. It re-ranks proximity and vertical-order candidates in separate family lists. The matched *joint P/V* control uses the same candidates, compatibility estimates, product scores, and support/contact positions and identities. It differs only by placing proximity and vertical-order candidates in one list. Table S12 reports this comparison at all five K values.

No rule is best in every setting. For Open3DSG, Joint P/V raises Linear Recall at all five K values. For the MLP, it raises Recall at $K = 5, 10$ but lowers Recall by 2.92 and 3.80 percentage points at $K = 50, 100$, with both paired intervals below zero. At Open3DSG $K = 50$, the MLP control changes the selected proximity/vertical-order counts from 6,295/6,508 to 3,423/9,380 while leaving all 13,833 support/contact selections unchanged. We therefore interpret family-aware re-ranking as a constraint that preserves relation-family composition, not as a rule that maximizes aggregate performance.

Two broader relaxations are retained as scope comparisons. *Support order only* preserves the relative source order of support/contact candidates but permits competition across families. *All-family product* also applies learned compatibility to support/contact. These conditions change support/contact positions or selected membership and are not matched tests of the family-aware constraint.

C.5 Paired Scan-Level Intervals

Paired 95% intervals use 1,000 scan-level bootstrap resamples. All contexts from a sampled scan are resampled together, and the same resample indices are used for Source and each RelCompat3D variant.

Table S13 reports the paired $K = 50$ changes used in the main-paper interval interpretation.

Predictor	Ranking rule	$K=5$	$K=10$	$K=20$	$K=50$	$K=100$
VL-SAT	Source	0.93	3.61	7.27	16.43	26.00
	Linear	0.51	3.06	6.36	14.34	21.50
	MLP	0.42	2.79	6.17	14.05	21.57
Open3DSG	Source	83.20	65.18	53.80	45.96	40.87
	Linear	1.26	1.60	2.90	4.90	8.65
	MLP	16.84	13.18	9.17	8.84	12.32
SGFN	Source	0.00	2.42	5.33	11.56	21.55
	Linear	0.00	1.54	3.89	8.13	15.98
	MLP	0.00	1.32	3.55	7.63	15.88

Table S8: Point- and mesh-based consistency audit. Agreement-based Violation (%, ↓) on the shared 3DSSG validation target. A satisfied or violated label requires agreement between the point- and mesh-based measurements.

		Five smooth mappings			Percentile mapping		
Predictor	Estimator	Pass	Min. ΔR	Max. ΔV	Pass	Min. ΔR	Max. ΔV
VL-SAT	Linear	25/25	0.000	−0.036	5/5	+0.025	−0.146
	MLP	24/25	−0.025	−0.109	3/5	−0.201	−0.146
Open3DSG	Linear	25/25	+0.302	−9.163	5/5	+0.277	−8.982
	MLP	25/25	+0.277	−8.240	5/5	+0.201	−7.851
SGFN	Linear	25/25	0.000	0.000	3/5	−0.227	0.000
	MLP	25/25	0.000	0.000	3/5	−0.050	0.000

Table S9: Source-score mapping sensitivity over $K \in \{5, 10, 20, 50, 100\}$. Pass counts settings with $\Delta R \geq 0$ and $\Delta V \leq 0$ relative to Source. Minimum Recall and maximum Violation changes are percentage points. Negative ΔV is preferred. The smooth block pools three power and two logit-temperature mappings.

Estimator	Predictor	ΔR@50	ΔV@50
Linear	VL-SAT	$+0.05$ $[-0.05, +0.18]$	-0.70 $[-0.88, -0.55]$
	Open3DSG	$+3.75$ $[+2.96, +4.50]$	-10.45 $[-11.06, -9.86]$
	SGFN	$+0.48$ $[+0.29, +0.69]$	-1.22 $[-1.44, -1.01]$
MLP	VL-SAT	0.00 $[-0.10, +0.10]$	-0.79 $[-0.98, -0.62]$
	Open3DSG	$+6.27$ $[+5.02, +7.49]$	-9.74 $[-10.31, -9.13]$
	SGFN	$+0.55$ $[+0.32, +0.81]$	-1.27 $[-1.51, -1.06]$

Table S13: Paired scan-level changes at $K = 50$. Each cell reports the RelCompat3D-minus-Source change and its paired 95% interval in percentage points. Positive Recall and negative Violation changes are favorable.

For Open3DSG and SGFN, both intervals exclude zero in the favorable direction for both metrics. For VL-SAT, the Recall intervals contain zero while the Violation intervals remain below zero.

C.6 Relation-Family Composition

Table S14 reports the slice for each family inside the actual global top-100 ranking. Deltas compare RelCompat3D-Linear with Source using shared paired context-bootstrap indices.

Predictor	Family	Selected	Share	ΔR	ΔV
VL-SAT	Support/contact	24,021	43.8%	.0000	.0000
VL-SAT	Proximity	16,513	30.1%	+.0040	−.0076
VL-SAT	Vertical order	14,266	26.0%	+.0051	−.0609
Open3DSG	Support/contact	27,162	51.8%	.0000	.0000
Open3DSG	Proximity	10,929	20.8%	+.0968	−.0443
Open3DSG	Vertical order	14,361	27.4%	+.1538	−.3014
SGFN	Support/contact	23,616	43.1%	.0000	.0000
SGFN	Proximity	9,562	17.4%	+.0142	−.0150
SGFN	Vertical order	21,622	39.5%	+.0051	−.0643

Table S14: Family composition and family-specific metrics within the selected top-100 predictions. Counts and shares are identical for Source, RelCompat3D-Linear, and RelCompat3D-MLP because family-aware re-ranking preserves the source family-label sequence. Deltas are RelCompat3D-Linear minus Source, and positive ΔV is a regression.

Support/contact is exactly unchanged because the ranking procedure preserves its selected subsequence and the source family-label sequence at every K. Vertical order drives most of the aggregate reduction. The result is consequently a scoped improvement, not a uniform family claim. The same direction holds across the reported K values.

Predictor	Ranking rule	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Source	41.94/0.29	63.22/0.82	80.74/1.42	92.72/2.68	96.35/4.76
	Robust density	41.14/0.33	62.13/0.97	79.83/1.51	91.77/2.68	96.10/4.46
	Linear	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	MLP	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
Open3DSG	Source	3.42/52.05	9.87/32.89	19.89/20.99	40.43/13.87	51.11/12.42
	Robust density	4.03/2.03	11.56/3.45	22.61/4.30	43.76/5.09	56.92/6.05
	Linear	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	MLP	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
SGFN	Source	31.17/2.37	39.75/3.49	49.12/3.22	74.02/3.85	92.35/6.30
	Robust density	31.14/2.37	39.43/3.58	48.51/3.31	73.87/4.01	91.84/6.19
	Linear	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	MLP	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50

Table S10: Comparison with the non-learned robust-density baseline on the shared 3DSSG validation scenes. Cells are exact-match Recall / verifier-derived Violation percentages. Linear and MLP denote the two proposed RelCompat3D estimators under the same family-aware re-ranking rule.

C.7 Verifier-Uncertainty Sensitivity

The reported V@K$_{all}$ includes uncertain candidates in its denominator but does not label them satisfied. We therefore recompute each ranking with V@K$_{dec}$ over satisfied/violated candidates, uncertainty rate $U@K$, and an uncertain-as-violation variant V@K$_{u\to v}$ that counts every uncertain candidate as a violation. Table S15 gives the $K = 100$ comparison.

Predictor	Ranking rule	Dec. cov.	V_{all}	V_{dec}	U	$V_{u\to v}$
VL-SAT	Source	.6536	.0476	.0729	.3464	.3941
VL-SAT	Linear	.6527	.0295	.0452	.3473	.3768
VL-SAT	MLP	.6542	.0296	.0453	.3458	.3755
Open3DSG	Source	.5839	.1242	.2126	.4161	.5402
Open3DSG	Linear	.5915	.0324	.0548	.4085	.4409
Open3DSG	MLP	.5735	.0371	.0648	.4265	.4637
SGFN	Source	.6268	.0630	.1005	.3732	.4362
SGFN	Linear	.6197	.0350	.0565	.3803	.4153
SGFN	MLP	.6219	.0350	.0563	.3781	.4131

Table S15: Primary-verifier status coverage and uncertainty sensitivity at $K = 100$. Status coverage is 1.0000 for every row. Decidable coverage is the satisfied or violated fraction among status-assigned candidates. U is the uncertain fraction, and $V_{u\to v}$ counts uncertain candidates as violations. Lower is better for V and U, while higher is better for coverage.

RelCompat3D-Linear-minus-Source changes in V_{dec} are $-.0277$ for VL-SAT, $-.1578$ for Open3DSG, and $-.0440$ for SGFN. Their paired 95% intervals all exclude zero. Uncertainty-rate changes are +.0009, $-.0076$, and +.0071. Uncertain-as-violation changes are $-.0173$, $-.0993$, and $-.0209$. Both decidable-only and uncertain-as-violation V are lower for all three predictors, although uncertainty does not decrease uniformly. Across both estimators, primary, decidable-only, and uncertain-as-violation V are non-increasing relative to Source in all 30 predictor–K settings. These denominator checks show that the Violation reduction is not specific to one treatment of uncertain candidates.

Predictor	Condition	R@50	V@50	R@100	V@100
Open3DSG	MLP full	46.70	4.13	59.89	3.71
	Wrong predicate	46.78	19.30	58.89	18.34
	Wrong pair	38.02	9.31	49.90	8.90
	Shuffled geometry	36.83	12.88	47.94	12.38
	Fixed-predicate swap	45.52	19.13	57.43	18.15
	Compatibility only	48.19	3.42	61.33	3.28
VL-SAT	MLP full	92.72	1.89	96.50	2.96
	Wrong predicate	90.99	5.03	94.99	8.31
	Wrong pair	91.31	2.62	95.62	4.71
	Shuffled geometry	90.26	3.00	94.71	5.49
	Fixed-predicate swap	90.99	4.99	95.09	8.13
	Compatibility only	77.84	1.61	90.18	2.05
SGFN	MLP full	74.57	2.58	92.88	3.50
	Wrong predicate	71.78	9.67	90.01	13.52
	Wrong pair	72.10	3.91	89.05	6.59
	Shuffled geometry	71.15	4.59	87.84	8.06
	Fixed-predicate swap	71.85	9.65	90.23	13.35
	Compatibility only	57.10	2.33	80.99	2.31

Table S11: Matched controls for RelCompat3D-MLP. Recall (R, ↑) and verifier-derived Violation (V, ↓) on the shared 3DSSG validation scenes. All values are percentages. Main Table 2 reports the corresponding Linear controls.

Predictor	Estimator	Re-ranking	$K = 5$	$K = 10$	$K = 20$	$K = 50$	$K = 100$
VL-SAT	Linear	Family-aware	42.07/0.15	63.39/0.57	80.82/1.14	92.77/1.97	96.58/2.95
	Linear	Joint P/V	42.04/0.15	63.42/0.55	80.84/1.11	92.72/1.95	96.60/2.94
	MLP	Family-aware	42.09/0.15	63.47/0.51	80.92/1.09	92.72/1.89	96.50/2.96
	MLP	Joint P/V	42.12/0.15	63.42/0.49	80.97/1.06	92.77/1.86	96.53/2.93
Open3DSG	Linear	Family-aware	3.73/0.94	11.38/2.33	23.62/3.13	44.18/3.42	56.85/3.24
	Linear	Joint P/V	7.98/0.94	15.84/2.33	25.33/3.13	44.76/3.39	57.83/3.08
	MLP	Family-aware	3.70/4.95	11.78/4.97	24.67/4.56	46.70/4.13	59.89/3.71
	MLP	Joint P/V	7.48/4.17	14.85/4.77	23.92/4.66	43.78/4.16	56.09/3.68
SGFN	Linear	Family-aware	31.17/2.37	39.75/3.47	49.14/2.97	74.50/2.63	93.03/3.50
	Linear	Joint P/V	31.17/2.37	39.75/3.47	49.19/2.97	74.67/2.61	93.28/3.39
	MLP	Family-aware	31.17/2.37	39.75/3.47	49.19/2.96	74.57/2.58	92.88/3.50
	MLP	Joint P/V	31.17/2.37	39.75/3.47	49.27/2.96	75.25/2.57	93.13/3.37

Table S12: Matched routing-constraint comparison on the shared 3DSSG validation scenes. Cells are exact-match Recall / verifier-derived Violation percentages. Joint P/V changes only the competition between proximity and vertical-order candidates. Support/contact positions and identities remain fixed.

D Secondary Diagnostics

These diagnostics bound sensitivity and scope. They are not additional main comparisons and do not broaden the claim beyond the shared 3DSSG validation scenes.

Across five fixed seeds, Linear is identical and MLP is stable. All predictor–K cells retain the source-relative direction except one VL-SAT run at $K = 50$, which selects one fewer exact relation and retains lower Violation. Eight one-at-a-time construction and objective perturbations change Linear by at most 0.35 Recall and 0.20 Violation percentage points and never reverse that direction.

A pooled Linear head produces predictor-dependent Recall–Violation trade-offs. Restoring the 15 omitted Open3DSG contexts changes $K = 100$ Linear Recall/Violation from 56.85/3.24 to 57.35/3.32.